Kien T. Huynh

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Education

University of California, Berkeley **Expected Graduation May 2027**
Bachelor's Degree - Computer Science *GPA: 3.88/4.00*

- **Fiat Lux Scholar:** Faculty-selected full-ride scholarship — 1 of 63 recipients in UC Berkeley's 2023 incoming class.
- **Relevant Coursework:** Data Structures, Efficient Algorithms, Operating Systems, Database Systems, Computer Networks, Compilers, Software Engineering, Computer Architecture, Computer Security, Machine Learning

Technical Skills

Languages: Python, TypeScript, C/C++, Ruby, JavaScript, SQL, OCaml, HTML/CSS
Frameworks: Ruby on Rails, React, Svelte, FastAPI, Flask, RSpec, Cucumber, Capybara
Data & ML: PyTorch, Pandas, Polars, PySpark, cuDF, NumPy, CVXPY
Tools & Platforms: GCP, AWS, Firebase, Docker, PostgreSQL, Node.js, Git, CI/CD (Actions), Render, Heroku, Sentry

Experience

CourseAssist AI **Incoming October 2026**
Software Engineer *Berkeley, CA*

UC Berkeley College of Engineering **May 2026 – Present**
Teaching Assistant, CS 169A: Software Engineering *Berkeley, CA*

- Led two weekly discussion sections and office hours across two terms (50 and 100 students), teaching Agile development, behavior- and test-driven development, MVC and SaaS architecture, design patterns, and refactoring of legacy code.
- Migrated the course deployment pipeline from Heroku to Render, authoring render.yaml blueprints that provision and bind a PostgreSQL instance per project and deploy-time hooks that run migrations and seeds without CLI access.
- Rebuilt Codio environment images on updated mise, Node, and Ruby toolchains; adapted the existing autograder to the new deployment target, and reworked the final team project codebase to run on it; and maintained staff repositories.

UC Berkeley Sky Computing Lab **August 2025 – December 2025**
Student Researcher, Data Systems & Compilers (PandaX) *Berkeley, CA*

- Developed a source-to-source transpiler converting Pandas workflows into Polars, PySpark, and cuDF, pairing AST-based rewrite rules with an agentic worker–verifier loop for API idioms that could not be programmed as static rules.
- Designed the verifier to re-execute translated blocks on sampled data and compare outputs and profiler-derived variable lifetimes against the original, iterating until semantics and dependence-graph properties were preserved.
- Benchmarked industry Pandas pipelines across different languages and backends on GCP VMs/GPUs, measuring 2–4× average speedups.

Projects

BJC Teacher Tracker | *Ruby on Rails, JavaScript, PostgreSQL, AWS, Sentry, Heroku* **January 2026 – Present**

- Shipped features across a Rails platform serving 1,000+ schools worldwide, as one of 4 developers in an Agile workflow using test-driven development and CI/CD; still contributing alongside the maintainer after release.
- Instrumented email deliverability tracking with Amazon SES over an SNS webhook, verifying message signatures and topic ARNs and handling hard and soft bounces separately to flag teacher records needing correction.
- Built Google Maps integration, paginated admin views, and school deduplication tooling that presents merge candidates for maintainer review; repaired CI failures from flaky Capybara specs and an outdated webpack build.

Neural Network from Scratch | *C++* **August 2026 – Present**

- Implemented a feedforward neural network in C++ with no external numerical libraries – writing the matrix class, linear algebra operations, activation functions, backpropagation, and gradient descent from first principles.
- Trained and evaluated on the MNIST handwritten digit dataset, reaching 97%+ test accuracy; extending the framework to convolutional, recurrent, and transformer architectures with attention implemented from scratch.

ActionMap | *Ruby on Rails, D3.js, PostgreSQL, Heroku* **June 2025 – August 2025**

- Integrated three external REST APIs (Geocodio, Congress, Google News) to resolve a user's address to their legislators and display related bills and news coverage; also added GitHub and Google OAuth authentication.
- Delivered an interactive D3.js National-to-State-to-County-to-Representatives map; deployed to Heroku.